SLIIT Department of Software Engineering - Faculty of Computing

Mobile Application Project: Farmora

Milestone 2: Interactive High-Fidelity Prototyping & Design System (LO3)

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1. Prototype Scope & Objectives

The primary objective of Milestone 2 (Lab 07) is to transform low-fidelity wireframes into an interactive, high-fidelity digital prototype adhering strictly to Material Design 3 (M3) and agricultural usability principles.

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2. Design System & Style Guide

2.1 Color Palette

- Primary Brand (Agri Green):** ‘#2E7D32‘ (Represents vitality, nature, harvest prosperity)
- Primary Container / Accent:** ‘#E8F5E9‘ (Soft contrast for card backgrounds)
- Secondary (Harvest Gold/Orange):** ‘#F57C00‘ (Highlights alerts, urgency, pending escrow)
- Neutral Dark (Text/Headers):** ‘#1E293B‘ (High legibility on sunlight screens)
- Neutral Light (Surface):** ‘#F8FAFC‘ (Clean, non-distracting backgrounds)
- Danger / Temperature Alert:** ‘#D32F2F‘ (Cold storage breaches, expired verification)

2.2 Typography

- Primary English/Latin:** Google Inter & Poppins (Weights: 400 Regular, 500 Medium, 600 SemiBold, 700 Bold)
- Sinhala Localization:** Noto Sans Sinhala (Full Unicode rendering support)
- Tamil Localization:** Noto Sans Tamil (Full Unicode rendering support)

2.3 Micro-interactions & Haptic Feedback

- Pull-to-refresh:** Custom leaf-sprout animation indicating live market price updates.
- Haptic Click:** Confirms barcode scanning and camera shutter actions in outdoor farm conditions.
- Dynamic Chips:** Status indicators for ‘Order Status‘ (‘Pending‘, ‘In-Transit‘, ‘Delivered‘, ‘Disputed‘).

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3. High-Fidelity Screen Architecture & Key Interactions

3.1 Onboarding & Multi-Role Authentication

- Seamless phone OTP login alongside biometric (Fingerprint/Face ID).
- Dedicated role switcher: **Farmer**, **Wholesale Buyer**, **Transporter /

Fleet Owner**, and **Agri Expert**.
- Role-specific color accents to avoid cognitive confusion when switching roles.

3.2 Farmer Hub & Smart Harvest Publishing

- Clean, stepped wizard for listing crops.
- Real-time video recording & Cloudinary video hosting integration.
- Barcode/QR generator rendering unique traceability hash ‘FARM-BATCH-<ID>‘.
- Live farm weather widget with micro-climate rain alerts.

3.3 Buyer Direct Marketplace

- Category carousel: *Leafy Greens*, *Root Crops*, *Fruits*, *Spices*, *Export Quality*.
- Product detail sheet featuring interactive video inspection: buyers preview harvest freshness recorded directly in the field before purchase.
- Escrow purchase breakdown: Commodity Price + Logistics Cost + Quality Guarantee Insurance.

3.4 Logistics Fleet Hub & Telematics

- Interactive Google Maps view rendering vehicle markers, depot pins, and transit polylines.
- Real-time Cold Storage Temperature Gauge (Target 2℃ - 7℃) with automated breach alert.
- Fleet manager bottom sheets to dynamically configure Branch Depots, Vehicles, and Drivers.

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4. Accessibility & Inclusive Design (WCAG 2.1 AA Compliance)

- Contrast Ratios:** Text-to-background contrast exceeds 4.8:1 across all primary screens for outdoor visibility in Sri Lankan sunlight.
- Minimum Tap Targets:** All buttons, FABs, and dropdown triggers comply with the 48x48dp minimum physical touch area.
- Screen Reader Support:** Semantic labels applied to all icon buttons and graphic status indicators.
- Multi-lingual Switcher:** Instant runtime switching between Sinhala, Tamil, and English with zero app reloads.